



Gosper curve

The **Gosper curve**, named after [Bill Gosper](#), also known as the **Peano-Gosper Curve**^[1] and the **flowsnake** (a spoonerism of snowflake), is a space-filling curve whose limit set is [rep-7](#). It is a [fractal curve](#) similar in its construction to the [dragon curve](#) and the [Hilbert curve](#).

The Gosper curve can also be used for efficient hierarchical hexagonal clustering and indexing.^[2]

Lindenmayer system

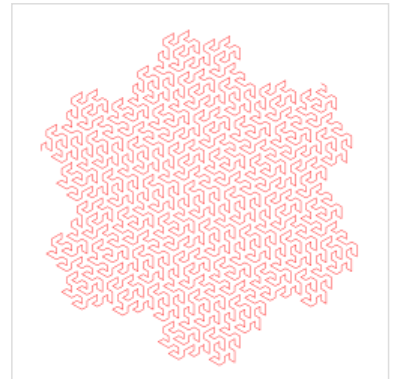
The Gosper curve can be represented using an [L-system](#) with rules as follows:

- Angle: 60°
- Axiom: *A*
- Replacement rules:
 - $A \mapsto A - B - -B + A + +AA + B -$
 - $B \mapsto +A - BB - -B - A + +A + B$

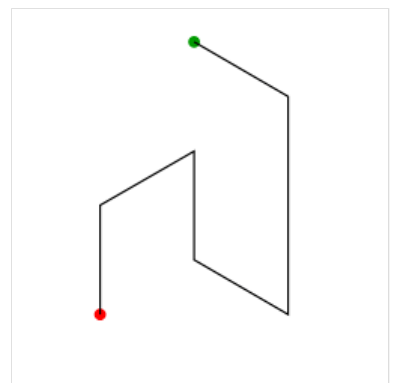
In this case both A and B mean to move forward, + means to turn left 60 degrees and - means to turn right 60 degrees - using a "turtle"-style program such as [Logo](#).

Properties

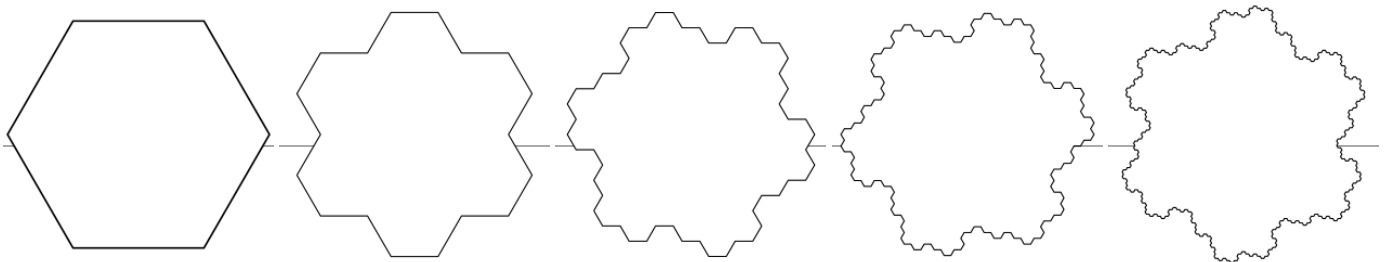
The space filled by the curve is called the **Gosper island**. The first few iterations of it are shown below:



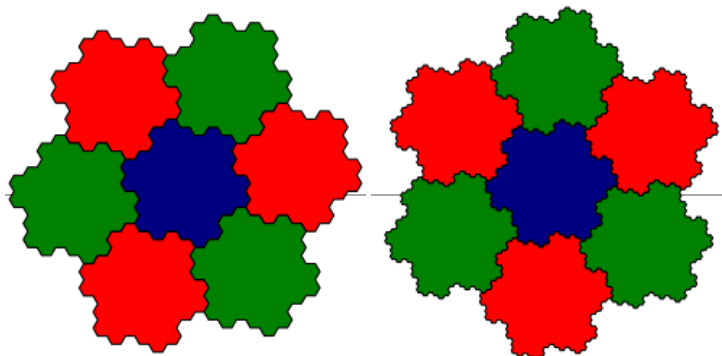
A fourth-stage Gosper curve



The line from the red to the green point shows a single step of the Gosper curve construction



The Gosper Island can [tile the plane](#). In fact, seven copies of the Gosper island can be joined to form a shape that is [similar](#), but scaled up by a factor of $\sqrt{7}$ in all dimensions. As can be seen from the diagram below, performing this operation with an intermediate iteration of the island leads to a scaled-up version of the next iteration. Repeating this process indefinitely produces a [tessellation](#) of the plane. The curve itself can likewise be extended to an infinite curve filling the whole plane.



See also

- [List of fractals by Hausdorff dimension](#)
- [M.C. Escher](#)

References

1. Weisstein, Eric W. "Peano-Gosper Curve" (<http://mathworld.wolfram.com/Peano-GosperCurve.html>). *MathWorld*. Retrieved 31 October 2013.
2. Uher, Vojtěch; Gajdoš, Petr; Snášel, Václav; Lai, Yu-Chi; Radecký, Michal (28 May 2019). "Hierarchical Hexagonal Clustering and Indexing" (<https://doi.org/10.3390%2Fsym11060731>). *Symmetry*. **11** (6): 731. Bibcode:2019Symm...11..731U (<https://ui.adsabs.harvard.edu/abs/2019Symm...11..731U>). doi:10.3390/sym11060731 (<https://doi.org/10.3390%2Fsym11060731>). hdl:10084/138899 (<https://hdl.handle.net/10084%2F138899>).

External links

- [NEW GOSPER SPACE FILLING CURVES](https://web.archive.org/web/20060112165112/http://kilin.u-shizuoka-ken.ac.jp/museum/gospere/343-024.pdf) (<https://web.archive.org/web/20060112165112/http://kilin.u-shizuoka-ken.ac.jp/museum/gospere/343-024.pdf>)
- [FRACTAL DE GOSPER](http://www.mathcurve.com/fractals/gosper/gosper.shtml) (<http://www.mathcurve.com/fractals/gosper/gosper.shtml>) (in French)
- [Gosper Island](http://mathworld.wolfram.com/GosperIsland.html) (<http://mathworld.wolfram.com/GosperIsland.html>) at Wolfram MathWorld
- [Flowsnake by R. William Gosper](https://larryriddle.agnesscott.org/ifs/ksnow/flowsnake.htm) (<https://larryriddle.agnesscott.org/ifs/ksnow/flowsnake.htm>)

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